

Invasions in the News:

Sentiment, Media Coverage and Signaling

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Abstract

How do media sources respond to significant disruptions in the international system? This study examines the reactions of state-controlled media in post-Soviet states and satellite countries to the Russian invasion of Ukraine, comparing them with independent media outlets. Utilizing an extensive corpus of several hundred thousand newspaper articles collected before and after the invasion on February 24, 2022, we employ a large language model trained on adversarially coded data to perform topic classification and sentiment analysis toward Russia. Our findings indicate that sentiment toward Russia becomes significantly more negative in independent media following the invasion, whereas the decline in sentiment is less pronounced in state-owned media. These results highlight the divergent roles of media types in shaping public opinion during international crises and contribute to our understanding of media behavior in times of geopolitical upheaval.

Introduction

How do media sources respond when there is a significant disruption to the status quo in international system? The Russian invasion of Ukraine in February 2022 was one of the most significant shocks to the international system in the post-Cold War era, the effects of which are still being parsed by researchers and politicians three years later. As

the North Atlantic Treaty Organization (NATO) debated how best to prepare for the possibility of an invasion of Ukraine (Jakes & Anderson, 2024) and a US presidential candidate openly encouraged Russia to invade NATO countries (Sanger, 2024), it is important for researchers to study how states are responding to the invasion with respect to their relationship with Russia. State media accounts form an important tool for states seeking to influence public opinion (Guriev & Treisman, 2020) and evidence suggests that selective attribution and sentiment play an important role in how news is displayed (Rozenas & Stukal, 2019). Furthermore, how elite cues impact public opinion is an important topic that has received significant attention within the international relations literature (Guisinger & Saunders, 2017). On the other hand, the effects of privately owned media on public opinion have received less attention; particularly when state and privately owned media diverge in sentiment. Most previous work has utilized survey experiments to examine the relationship between public opinion and media rather than exploring actual cues from elites. Understanding how strategic elites utilize media messaging to respond to significant events in the international sphere is an important next step in identifying the structure of the relationship between the public and political elites.

Signalling is another important topic in international relations literature. Much of the theoretical work has centered on how to signal credibility and capability to possible rivals (Fearon, 1997). Other work has suggested that allies may signal by placing troops abroad, prepared and ready to defend alliance commitments (Blankenship & Lin-Greenberg, 2022). In this article, we argue that the role of state-owned media as a signalling device has been overlooked in the literature. We argue that state’s strategically control the sentiment expressed towards Russia in their media, signalling a lack of condemnation of Russia’s actions. This stands in stark contrast with independent media which, unconstrained by these strategic considerations, is free to engage in harsh criticism of Russia’s actions.

Previous work has suffered from a lack of access to the data necessary to examine these dynamics at scale, but advances in natural language processing have opened

up new avenues for research into the topic. To examine this phenomenon within the real world, we studied how state controlled media in former Soviet states as well as satellite and allied states responds to the Russian invasion of Ukraine compared to independent media sources. To do so, we used an extensive corpus of several hundred thousand newspaper articles collected in the period before and after the invasion on February 24th, 2022. We applied a keyword based approach to identify articles about Russia and then utilized sentiment analysis to examine how the overall sentiment of these articles changes around the discontinuity of the invasion. This approach will allow us to causally identify how strategic elites responded to the crisis situation introduced by the invasion and shed further light on the nature of the relationship between elites and the public as well as the role of public opinion in foreign policy.

We find that independent media responds significantly negatively to the Russian invasion of Ukraine. Sentiment towards Russia declines by 0.17 points on a scale of -1 (negative) to 1 (positive), a change of 0.27 standard deviations. Conversely, state-owned media responds significantly less negatively, to the tune of 0.06 points less negative, or 0.10 standard deviations. We argue that the explanation for this difference is strategic. Small states that are both geographically and geo-politically close to Russia utilize state-owned reporting to influence public opinion but also to signal their lack of opposition to the Russian invasion. In support of this argument, we report that the comparative difference in decline in sentiment is true even in non-military related articles, suggesting that while independent media engages in widespread condemnation of Russia, state-owned media's response is significantly more conservative.

The article proceeds as follows. We first identify relevant literature on alliance politics and signalling as well as the role of the media. We then discuss the case at hand, with a focus on the Russian invasion of Ukraine as well as which states were used in our analysis. We highlight their relationship to Russia today as part of our review. In our third section, we discuss our theory, identifying hypotheses concerning the role of the media based on foreign policy alignment with Russia, as well as the role of state-owned and privately owned media. In our fourth section, we discuss data,

research design and our results. The conclusion assesses next steps as well as the implications of divergent media on public opinion.

Literature Review and Theory

Intuitively, the idea that state and independent media might react differently to a violation of an international border is not a novel concept. The two media sources might have very different mandates as well as funding sources. In addition, as a tool of the state, state-owned media is restricted by the states' decisions on how to make and signal alliance commitments. The international relations literature has grappled with several important questions related to alliances. Do states align with those most powerful and closest to them? Do states keep their friends closer, and their enemies closer? Upon deciding who to align with, how can small states, without powerful weapons and additional troops to send abroad, signal their commitment? The lack of concrete theory to help explain this variation in media response means that we must look to alliance, signaling, and media literature to help us begin to understand why we might see a difference in media sentiment.

Weaker states often choose to balance in an effort to maintain their place in the system (Waltz, 2010). When states do this, they are choosing the weaker side due to the perceived threat of the stronger state. States might also choose to bandwagon, that is join the stronger side (Walt, 2007; Waltz, 2010). States may choose to align with those who have the most aggregate offensive capabilities, who are located most closely, and who have the most aggressive perceived intentions. While bandwagoning might seem more attractive, Walt suggests that balancing is more common among states that matter and who have more capabilities (Walt, 2007).

While security concerns are likely to drive alliance preferences, states may also choose to join alliances that more closely adhere to their state views. Ideologically similar alliances are more common than alliances among countries of opposing ideological systems (Walt, 2007). However, politicians might use ideological similarities to tout the strength of an alliance, though this level of closeness might be a facade,

designed to inspire feelings of in-group favoritism.

States may also enter into alliances designed to constrain. While some alliances are capability aggregation measures, meant to stand firm against external threats, alliances also may serve to tie squabbling partners together (Weitsman, 2004). The alliance paradox suggests that allies are not only frequently unreliable, but also prone to confrontation. When states tether themselves by aligning with a powerful ally, they facilitate communication and mitigate potential hostilities, but this strategy constrains their partners. Tethering happens most often in relatively higher threat relationships; states that see themselves threatened by low levels of threat are more likely to hedge. States that are less threatened will "allow the hedging state to draw others into its sphere of influence as long as the course of action is not overly costly or too provocative to potential enemies" (Weitsman, 2004). Small states located near powerful, threatening states might therefore choose to tether themselves to the regional pole, hoping to remain sovereign.

Once states have decided on the best partners, how can they signal that they are committed? Signalling might involve words or deeds (Fuhrmann & Sechser, 2014). Most political science literature focuses on how states might signal to rival states their level of capability and resolve (Fearon, 1997; Kertzer, 2016). Others have noted that signalling to allies is particularly difficult (Schelling, 1967). Committed allies might make public statements or they might deploy troops or weapons to allied locations (Blankenship & Lin-Greenberg, 2022; Fuhrmann & Sechser, 2014).

By making public statements, countries could signal to their allies that they are committed to maintaining alliances. One of the most effective ways to broadcast these statements might be through the use of the media, including television, newspapers, and social media. This allows these statements to be public, helping to tie hands (Fearon, 1997). The news media is often used to set agendas and prime audiences as well as signal elite cues (Chong & Druckman, 2007; Iyengar & Kinder, 2010). The news media can set agendas by identifying and publicizing issues for the mass public. The media can also prime by influencing the issues that the public pays attention

to when they make judgements. Importantly, the media can help to transmit elite signals. There are two general types of media in most states: state-owned and private, or independent media.

The role of state-owned and private media firms is unclear in the literature. Public choice theory suggests that government-owned media outlets “distort and manipulate information” to prop up the incumbent regimes, while privately owned and independent media sources offer alternative views with largely unbiased and accurate information (Djankov et al., 2003). State-owned media might also be more closely aligned with past authoritarian regimes (Moehler & Singh, 2011). On the other hand, state-owned firms might provide more fair and balanced accounts, given their governmental affiliation and monetary backing (Djankov et al., 2003). Independent media, unhindered by government regulation may be more free to seek out the full story, or they may be backed by private firms that seek to provide a narrative more aligned with a desired end state. Privately owned media outlets might be more responsive to the public and open to opposing perspectives (Moehler & Singh, 2011). Independent media in states that are more fully aligned with western values, with journalists interested in maintaining the freedom of the press that comes as a part of the international liberal order should be more likely to report more accurately. This will help them to maintain credibility as well as demonstrate their commitment to showing what is going on in the world.

Given this literature on the alignment of independent media with western, liberal values, we expect that independent media will respond negatively to Russia’s invasion of Ukraine. If this is the case, we would expect that sentiment will decline severely around the discontinuity of the invasion:

Hypothesis 1: Given the international liberal order, independent news sources should respond negatively toward Russia after the invasion of Ukraine.

However, it might be the case that rather than reporting more negative sentiment towards Russia, coverage simply shifts towards more reporting on the war itself, and military topics more generally. If this is the case, and these topics are more frequently

coded as a negative, then our analysis might actually be capturing a change in the substance of coverage, rather than a change in sentiment. To support our hypothesis that sentiment towards Russia will become more negative, we propose hypothesis 1A:

Hypothesis 1A: Even in topics unrelated to military matters, given the international liberal order, independent news sources should respond negatively toward Russia after the invasion of Ukraine.

While the independent media might have more freedom to engage in critiques of Russia, state-owned media might be more constrained. Given the nature of the relationships between the states in this analysis and Russia, we expect that state-owned media will strategically manage reporting after the onset of the invasion. We hypothesize that states that are most closely aligned with Russia will display different preferences in their state-owned and private media. Smaller nations aligned more closely with Russia located in the Russian near abroad might use their official media sources to signal support for Russian actions. By signalling through official words, small states might be demonstrating their tethers to Russia, decreasing the chances that Russia will seek to invade their sovereign territory. This leads to hypothesis 2:

Hypothesis 2: State-owned media will report negative sentiment towards Russia after the invasion to a lesser extent than independent media.

However, it is possible that some level of strategic reporting will concern topic coverage rather than control of sentiment. It is plausible that independent media might report on the war more than state-owned media, and if coverage of the war is in general negative, then what we would capture in hypothesis 2 is not a difference in sentiment, but rather a difference in coverage. Therefore, we propose hypothesis 2A:

Hypothesis 2A: Even in topics unrelated to military matters, state-owned media and independent media will display different sentiments towards Russia after the invasion of Ukraine.

The Case

The dissolution of the Soviet Union in 1991 was a significant disruption to the international system. Not just unexpected, it was also peaceful, moving the world from a bipolar one to a unipolar one (Monteiro, 2014; Waltz, 2010). This abrupt transition surprised academics and policymakers alike. While academics do not have a well-developed theory accounting for change in the structure of the international system, most attribute alterations to conflict and the settlements after war (Ikenberry, 2001; Waltz, 2010). The lack of a competitor allowed for the U.S. led liberal world order to flourish, with former client states of the Soviet Union largely encouraged to join institutions that allowed for greater cooperation and prosperity. Many states also adopted the trappings of democracy, with elections and nominal freedoms granted to their citizens.

In the case of Russia, in the years after the fall of the Soviet Union, the state held elections, allowed for some level of press openness, and made overtures towards continued cooperation with the west. The states that allied with Russia also sought to build democratic institutions, though these were often thin institutions and sometimes failed to provide the true freedoms of the press and speech.

Many former Soviet states maintained close ties with Russia. Other former allies of the Soviet Union often chose new partners, with many realigning their politics to look toward the West. Some sought to join the North Atlantic Treaty Organization (NATO) and the European Union (EU). Russia even joined Partnership for Peace, signalling a commitment to security cooperation and global institutions.

The twelve countries in our dataset are all in relative proximity to Russia. Even prior to the 2022 invasion of Ukraine, Russia’s ability to project power into to allies around the world was well-known, with Russia supporting efforts in Syria, South Africa, and Venezuela (Charap et al., 2019; Radin et al., 2019; “US-Russian spat over bombers landing in Venezuela”, 2018). The Russians also maintained the ability to project greater volumes of military power into its near abroad. Knowing this, states must

make choices: do they bandwagon with Russia or do they choose to balance against the state?

Seven of the twelve countries in our dataset, Armenia, Azerbaijan, Belarus, Georgia, Kazakhstan, Moldova, and Ukraine all became independent states after having been Republics of the Soviet Union (“The Belarus-Russia Alliance”, n.d.). With the exception of Ukraine, the former Soviet states joined the Commonwealth of Independent States (CIS) in 1993. The CIS encourages cooperation in economic, political, and military matters (Nations, n.d.). After the Russo-Georgian War in 2008, Georgia left the CIS (Team, 2006). Though Ukraine had refused to formally join the organization, the country had continued to participate until 2018 (“Ukraine Announces Plans To Quit CIS, Terminate Parts Of Russia Friendship Treaty”, 2018). Additionally, Moldova has recently announced its intention to withdraw by the end of 2024, due to territorial incursions by Russia (Chiu, 2023).

Former allies of the Soviet Union also reconsidered their foreign relations. Albania and Hungary, two former Soviet Allies, both joined NATO, choosing Western alliances (NATO, n.d.). While Hungary officially joined the European Union (EU), Albania is still working toward accession negotiations. Yugoslavia, a former ally of the Soviet Union, saw a brutal war in the 1990s. Serbia is a former constituent republic of Yugoslavia, forming a new federation with Montenegro in 1992, ultimately declaring independence in 2006. Kosovo, a former autonomous province within Yugoslavia, currently has partial diplomatic recognition with 104 member states of the UN recognizing it as a sovereign state. For the purposes of this project, we will treat Kosovo as a full state (“Russia’s Influence in the Balkans”, n.d.). Finally, Turkey was an important member of the anti-Soviet bloc during the Cold War. A member of NATO since 1951, Turkey has since cultivated closer ties with Russia, normalizing relations after 2016 and the Syrian crisis. Turkey did not join the rest of the West in imposing sanctions on Russia, though President Recep Tayyip Erdoğan also vowed not to accept territorial changes made through the use of force.

All twelve states, and importantly, Russia, have acceded to the United Nations

Charter, which explicitly guarantees that "all Members shall refrain in their international relations from the threat or use of force against the territorial integrity or political independence of any state" (Nations, n.d.). We should thus expect that Russia, as well as all successor states and former allies would respect state sovereignty and refrain from using force to change the facts on the ground of other states.

While Russia succeeded the Soviet Union seat in the the UN, over the past twenty years, Russia has also assisted South Ossetia and Abkhazia in their pursuit of independence from Georgia and pursued claimed "de-Nazification" programs in nearby Ukraine. Observers have criticized these efforts as unwarranted aggression in Russia's near abroad. Russia's 2014 annexation of Crimea occurred after political upheaval in Ukraine and despite efforts to end the conflict, including the Minsk Agreements, brokered by Western nations, violence has only escalated. The Minsk Agreements called for an immediate ceasefire as well as the restoration of state borders, while reaffirming the sovereignty of the state of Ukraine. Ostensibly, Russia invaded Ukraine to support pro-Russia separatists fighting for independent states (Walker, 2023). A referendum held after the annexation reportedly had 95.5 percent of voters supporting a union with Russia. However, these results are not internationally recognized. The use of democratic concepts, including allowing citizens to vote on state matters, suggests that Russia understands the power of appearing to adhere to the international order.

Though Russian leader President Vladimir Putin claimed that the results suggested that the people of Crimea sought self-determination, much of the world condemned his actions. The leaders of the G7, including Canada, France, Germany, Italy, Japan, the United Kingdom and the United States, condemned the "Russia's continued actions to undermine Ukraine's sovereignty" ("Russia and Ukraine", n.d.). Additionally, the G7 leaders placed coordinated sanctions on Russia, imposing travel bans as well as freezing funds. The UN General Assembly condemned the invasion and subsequent referendum, with a vote of 100 to 11 against recognizing the results of the voting (Gumuchian, 2014). The United Kingdom Foreign Secretary suggested that "the result reinforces the fundamental principles upon which the UN was founded: principles of territorial

integrity and of the non-use of force” (Walker, 2023).

Between 2014 and 2021, the international community continued to support Ukraine’s bid for sovereignty. Ukraine signed a Comprehensive Assistance Package with NATO and voted to restore NATO membership as a strategic foreign policy objective. Additionally, the country signed an association agreement with the EU. These actions suggest that Ukraine’s desired foreign policy alignment is with the West.

In November of 2021, Russia mobilized troops along the Ukrainian border and Ukrainian President Zelenskyy continued to press for commitments from NATO for membership (Walker, 2023). Though President Joe Biden declined to provide this assurance from NATO, in a December 7, 2021 conversation with Russian President Putin, he did warn Putin of sweeping Western economic sanctions if an invasion occurred. The G7 also issued a statement, warning of Russian aggression, calling on Russia to “deescalate, pursue diplomatic channels, and abide by its international commitments on transparency of military activities” (“Russia and Ukraine”, n.d.).

In response to this public condemnation, Russia demanded that NATO remove troops and weapons from states that joined post 1997, including the Baltic states and Poland. Russia also demanded that NATO admit no new states, including Ukraine and Georgia, signalling that while some states were allowed to choose their post-Soviet alliances, others would be more constrained. Both the U.S. and NATO rejected the proposal, though the U.S. offered to participate in arms control negotiations. NATO Secretary-General Jens Stoltenberg also called for deescalation and dialogue, with the option of formal ties between Russia and NATO to avoid additional fighting (“US and NATO deliver responses to Russian demands over Ukraine”, n.d.). While Russia initially suggested that the government would study the proposal, both the U.S. and U.K. announced that they had intelligence suggesting that a Russian invasion of Ukraine was imminent.

On February 10, 2022, Russia launched massive military exercises with Belarus close to the Ukrainian border. These exercises were touted as practice for “suppressing and repelling external aggression during a defensive operation” (Walker, 2024). Instead

of the reported 30,000 personnel participating, western intelligence sources reported that up to 140,000 troops were located on the border with Ukraine (Walker, 2024). Even with ongoing Western pressure, Russian president Putin continued to exercise his troops, including his strategic nuclear forces, testing a variety of hypersonic and cruise missiles.

Finally, on 24 February 2022, President Putin announced that Russian forces would carry out a special military operation in Ukraine. The incursion began with missile strikes and just days later, fighting starts in Kyiv. Putin highlighted the threat that the need to “de-Nazify” and “de-militarize” Ukraine (ukraine backgrounder/conflict cross-roads). The international community largely condemned the attack, with the Council of Europe suspending Russian participation. The UNSC rejected a draft resolution, put forth by Albania, condemning the attack, because Russia vetoed the measure (Walker, 2024). The war continues today, with both sides launching multiple offensives to take and retake ground.

The Data

To study how the invasion changes sentiment towards Russia in state-sponsored media versus independent media, we utilized a corpus of news articles collected as part of [Machine Learning for Peace Project](#). These articles were collected via an extensive scraping process that targets both international and domestic news sources around the world. As part of the collection process, we traced ownership of each newspaper to identify those which are either directly state-owned or are indirectly captured by the state. Both independent and state-owned newspapers were carefully scraped via custom parsers in their original language and then human coders checked each newspaper for errors in the collection process. We then utilized a system of [Hugging Face Transformers](#) to translate articles into English. These translations have undergone human validation to ensure accuracy and usability. Once articles have been translated, we utilized a carefully curated keyword list of over 500 words associated with Russia,

including prominent politicians and companies, to identify articles related to Russia.¹

We collected articles from the 6 months on either side of the invasion, spanning from August 1st, 2021 till August 30th, 2022. Our curation process yielded 225,000 articles from 12 countries² and 70 sources, including at least one state-owned media source for each country. We chose these 12 countries according to two primary criteria. First, we chose a selection of former USSR and USSR-aligned states. Second, to ensure that Hungary, as a current NATO ally, was not unique, we added Turkey and Albania. Finally, we included Serbia as a non-NATO, non-former-USSR country, but with a government that has been decidedly pro-Russian in recent history to expand the analysis. For our main specifications in this article, we omit Ukraine to highlight the behavior in countries not directly affected by the invasion, but the inclusion of Ukraine does not meaningfully change the results.³ After omitting Ukraine, we were left with 174,000 articles from 62 sources.

To conduct our analysis, we utilized a zero-shot GPT 4.0 Turbo model to code articles across three dimensions; article topic (1 of 5), whether the article features Russia as a main actor (binary), and the sentiment of the article towards Russia (ternary). This coding was applied to the article title as well as the first two sentences.⁴ Russia was considered to encompass both the broader state, as well as features of the state apparatus including the Russian military and ministers of the Russian government as well as the head of the state, Vladimir Putin. The topics were classified as one of “Political,” “Economic,” “Military,” “Culture,” or “Other.” Sentiment was allowed to take the values of -1 (negative), 0 (neutral) or 1 (positive). To ensure accuracy, we compared the model’s performance against a test set of 100 randomly selected human-coded articles. The human coding process was adversarial double-coded by the coauthors who then debated the codings to settle on a final coding for each dimension

¹For more information on the process of article collection, please see [the MLP Pipeline Report](#).

²Albania, Azerbaijan, Armenia, Belarus, Hungary, Kazakhstan, Kosovo, Moldova, Serbia, Turkey, Ukraine, and Georgia

³To be included in future appendix.

⁴Unlike the GPT classifier, the human codings utilized both the title as well as the whole article. There were not significant performance gains from applying the GPT classifier to the whole article, and thus the classifier was applied to a truncated portion of the article to save both computational and monetary costs.

for each article. The overall agreement rate for initial human-codings was 79% with respect to sentiment, 91% with respect to Russian-Related, and 93% for topic. The final results from the GPT model with respect to sentiment are presented in Figure 1. Overall, the F1 score for the sentiment classifier is .92, for the Russian involvement classifier is .83, and for the Topic classifier is .80, with the confusion matrices reported in the appendix. The classifier raises two areas of concern. First, it is overly likely to classify political articles as other compared to human-coders. Second, it is overly likely to classify articles as not having Russian involvement compared to human-coders. Neither are likely to bias the findings reported in this article as any bias resulting from either issue would be towards a null result.



Empirical Analysis

We leverage two types of methods to test our hypotheses. First, we use a regression discontinuity design (RDD) to show that the sentiment of independent media coverage towards Russia becomes significantly more negative after the invasion (hypothesis 1). This increase in negative sentiment extends beyond articles related directly to the military (hypothesis 1A). Second, we use a difference-in-discontinuities (diff-in-discs) design (Grembi et al., 2016), (Toral, 2024) to show that the invasion causes a differential effect on sentiment in state-owned media as opposed to independent media (hypothesis 2). This difference in sentiment is pronounced even in non-military coverage (hypothesis 2A). Together, these two sets of causally identified evidence demonstrate that the invasion has a negative impact on sentiment towards Russia, but that state-owned media strategically tailors its coverage of Russia post-invasion.

The Invasion Makes Independent Media Sentiment Towards Russia More Negative: Regression Discontinuity Evidence

To test whether independent media sentiment towards Russia becomes more negative as a result of the invasion, we leverage an RDD around the day of the invasion. We find that the invasion decreases sentiment towards Russia by .27 standard deviations ($p < .001$). Similarly, for articles that were not classified as military related, we find that invasion decreases sentiment towards Russia by .25 standard deviations ($p < .001$).

Design

We exploit the discontinuity around the date of the invasion (February 24th, 2022) to measure the causal effect of the invasion on sentiment towards Russia expressed in independent news articles from our 11 countries. The key assumption of this design is that potential outcomes are continuous around the invasion. We believe this assumption is reasonable given the nature of time as a forcing variable in the context of article publications. Formally, treatment T is assigned by the difference between the date of article publication (D_p) and the date of the invasion (D_I) according to:

$$T = \begin{cases} 1 & \text{if } D_p \geq D_I \\ 0 & \text{if } D_p < D_I \end{cases} \quad (1)$$

The estimand of interest is $\gamma = \mathbb{E}[Y_p(1) - Y_p(0)]$, where $Y_p(1)$ and $Y_p(0)$ represent the potential outcome of interest (sentiment towards Russia) post-invasion and pre-invasion, respectively. We can identify the local average treatment effect (LATE) around the cutoff of the invasion by taking the difference in means before and after the invasion threshold:

$$\widehat{\gamma_{rdd}} = \lim_{D_P \downarrow D_I} \mathbb{E}[Y_p(1)|D_P = D_I] - \lim_{D_P \uparrow D_I} \mathbb{E}[Y_p(0)|D_P = D_I] \quad (2)$$

Estimation and Inference

We use a local linear regression applied to the following estimating equation:

$$Y_p = \beta_0 + \beta_1 T_p + \beta_2 D_p + \beta_3 T_p D_p + \alpha + \varepsilon_p \quad (3)$$

Where Y_p indicates the sentiment of articles published on date p and ε_p is the error term. α is a series of fixed effect estimators, including country, source, and topic fixed effects. If the RDD assumptions hold, then the LATE in Equation 2 is identified by β_1 . For all specifications, we use HC1 heteroskedasticity-robust standard errors clustered at the source level. For robustness, we followed Toral, 2024 and also tested a local linear regression with a triangular kernel (Cattaneo et al., 2024) within the optimal bandwidth (Calonico et al., 2020). This has no substantive impact on the results.

To analyze hypothesis 1A, that sentiment towards Russia would decline even in articles unrelated to military topics, we utilized the same analyses, but subset the articles to exclude those that our GPT model had identified the topic as military.

Table 1: Difference in Difference Analysis

	(1)	(2)	(3)	(4)	(5)	(6)
Invasion	-0.168*** (0.012)	-0.158*** (0.011)	-0.157*** (0.011)	-0.1500*** (0.0099)	-0.173*** (0.014)	-0.157*** (0.014)
State FEs	-	X	X	X	X	X
Source FEs	-	-	X	X	X	X
Topic FEs	-	-	-	X	X	X
Num.Obs.	135356	135356	135356	135356	43907	94983
R2	0.025	0.040	0.050	0.074	0.065	0.083
R2 Adj.	0.025	0.040	0.049	0.074	0.064	0.082

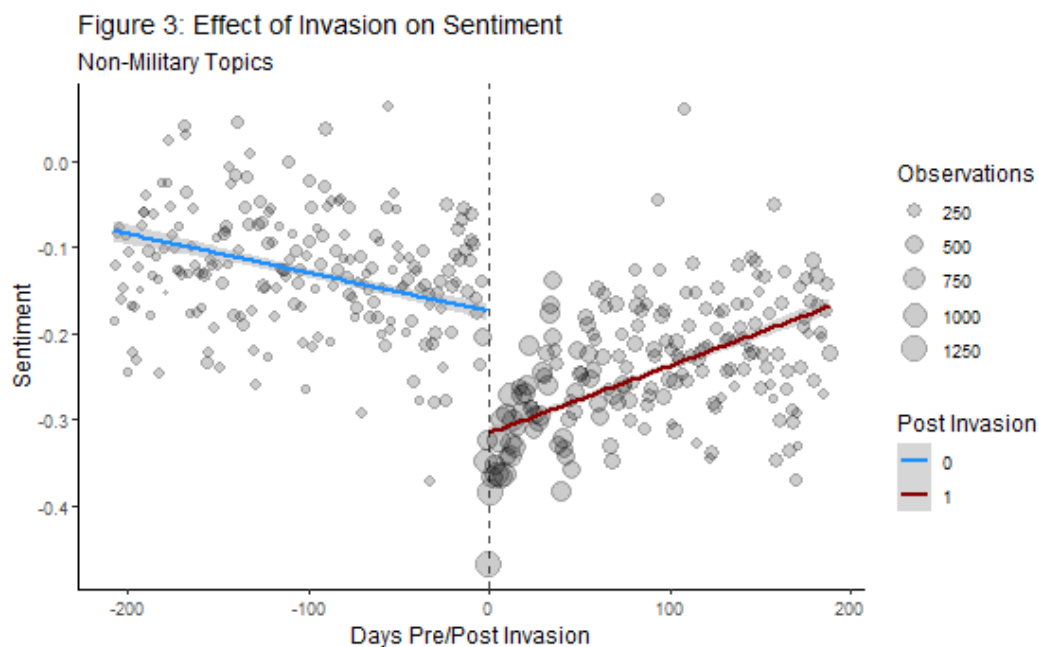
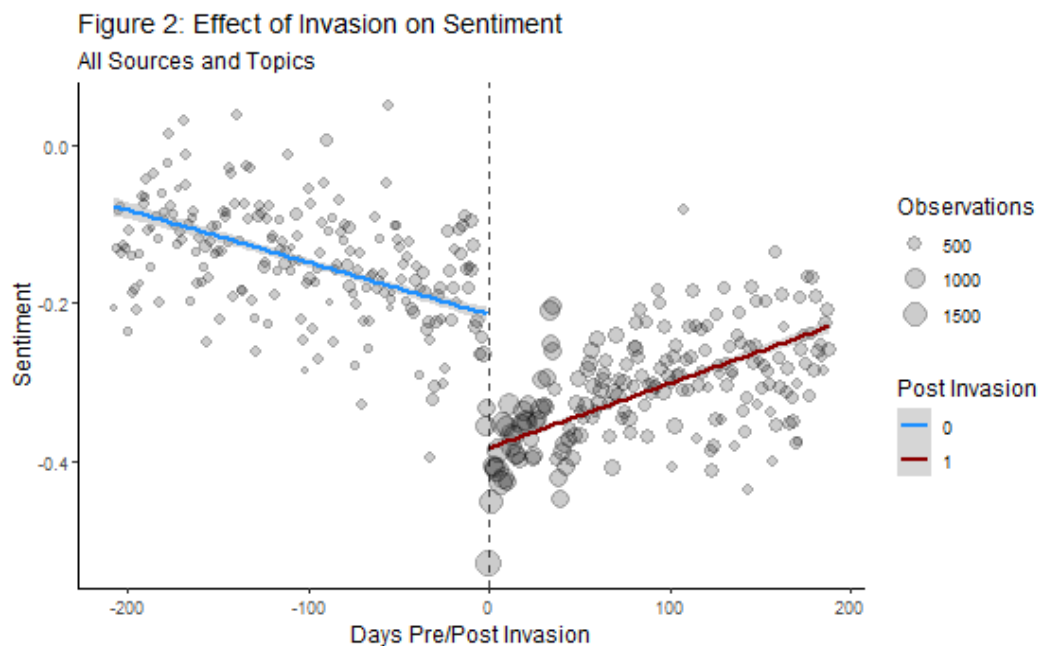
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Model 5 uses the triangular kernel within the optimal bandwidth. Model 6 omits articles that were coded as military-related.

Results

Table 1 presents our RDD results. Model 1 demonstrates that sentiment towards Russia in independent media became significantly more negative in response to the invasion of Ukraine. This represents a decline in sentiment of 0.17, which is a 0.27 standard deviation decline. Substantively, this means that the mean sentiment of articles fell from -0.17 pre-invasion to -0.32 post-invasion. Models 2, 3, and 4 show that this result is robust to the inclusion of state, source and topic fixed effects. Model 5 shows the results limited to the triangular kernel within optimal bandwidth, which do not change the interpretation of the findings. Finally, model 6 shows support for hypothesis 1A, that the effects of the invasion on sentiment are not limited to just articles related to military topics, but are true across our other four topics.

Figure 2 shows the effect of the invasion utilizing a linear model specification. Appendix Figure 2 shows that the model is immune to non-linear specifications. Figure 3 shows the effect of the invasion on sentiment for non-military topics.



In summary, these results demonstrate that independent media sources responded to the Russian invasion of Ukraine by significantly increasing negative sentiment towards Russia. This is true even in articles that do not directly cover military-related topics. This suggests that independent media, even in countries that are predisposed to be friendly toward Russia viewed the invasion negatively.

State-owned Media Responds Less Negatively to the Invasion than does Independent Media: Difference-in-Discontinuities Evidence

If state-owned media strategically controlled sentiment towards Russia in its coverage following the invasion of Ukraine, then the invasion should have a differential impact on article sentiment for independent sources and state-owned sources. If our theory is correct, independent media sentiment towards Russia should be significantly more negative after the invasion than the sentiment expressed in state-owned media.

To test whether state-owned media responds less negatively to the invasion than independent media, we leverage a diff-in-discs around the day of the invasion. Following (Toral, 2024) and (Grembi et al., 2016), this design combines a classic difference-in-differences (comparing state-owned articles and independent media articles) analysis with a regression discontinuity (comparing the sentiment towards Russia pre-invasion and post-invasion). We find that the invasion decreases sentiment towards Russia by 0.27 standard deviations ($p < .001$) for all articles, but that this effect is 0.10 standard deviations smaller for state-owned media sources ($p < .01$). Similarly, for articles that were not classified as military related, we find that invasion decreases sentiment towards Russia by 0.19 standard deviations ($p < .001$) for all articles, but that this effect is 0.11 standard deviations smaller for state-owned media sources ($p < .01$).

Design

This design exploits two factors. First, we exploit whether a source is state-owned or independent (called O_p) and second, we exploit the discontinuity around the date of the invasion (February 24th, 2022) (called D_p). This allows us to measure the causal effect of the invasion on sentiment towards Russia expressed in independent news articles from our 11 countries compared to the effect on sentiment in state-owned media. Formally, treatment T is assigned by the difference between the date of article publication (D_p) and the date of the invasion (D_I) according to:

$$T = \begin{cases} 1 & \text{if } D_p \geq D_I \\ 0 & \text{if } D_p < D_I \end{cases} \quad (4)$$

However, to separate the effect of whether an article is published by an independent media source or a state-owned media source, O_p is assigned according to:

$$O_p = \begin{cases} 1 & \text{if the source is state-owned} \\ 0 & \text{if the source is independent} \end{cases} \quad (5)$$

Potential outcomes, therefore, are a function of both O_p and D_p . We define the potential outcome as $Y_p(o, d)$. With that notation, the estimand of interest is:

$$\begin{aligned} \gamma = & \mathbb{E}[Y_p(1, 1) - Y_p(0, 1) | D_p = D_I, O_p = 1] \\ & - \mathbb{E}[Y_p(1, 0) - Y_p(0, 0) | D_p = D_I, O_p = 0] \end{aligned} \quad (6)$$

We can identify the local average treatment effect (LATE) around the cutoff of the invasion by taking the difference in means before and after the invasion threshold:

$$\begin{aligned} \widehat{\gamma_{ddisc}} = & \left(\lim_{D_P \downarrow D_I} \mathbb{E}[Y_p(1, 1) | D_P = D_I, O_p = 1] \right. \\ & - \lim_{D_P \uparrow D_I} \mathbb{E}[Y_p(0, 1) | D_P = D_I, O_p = 1] \Big) \\ & - \left(\lim_{D_P \downarrow D_I} \mathbb{E}[Y_p(1, 0) | D_P = D_I, O_p = 0] \right. \\ & \left. - \lim_{D_P \uparrow D_I} \mathbb{E}[Y_p(0, 0) | D_P = D_I, O_p = 0] \right) \end{aligned} \quad (7)$$

This design relies on two assumptions (Grembi et al., 2016). First, potential outcomes are continuous around the invasion. As in the previous section, we believe this assumption to be reasonable given the nature of the forcing variable. Second, that

there are parallel trends for state-owned media sources and independent sources. Figure 4 shows that the parallel trends assumption is a reasonable one. With these two assumptions, the diff-in-disc estimates the heterogeneous LATE (HLATE) causal effect of the invasion on sentiment, close to the threshold, for state-owned media sources and independent media sources.

Estimation and Inference

We use a local linear regression applied to the following estimating equation:

$$Y_p = \beta_0 + \beta_1 T_p + \beta_2 D_p + \beta_3 T_p D_p + O_p(\theta_0 + \theta_1 T_p + \theta_2 D_p + \theta_3 T_p D_p) + \alpha + \varepsilon_p \quad (8)$$

Where Y_p indicates the sentiment of articles published on date p and ε_p is the error term. α is a series of fixed effect estimators, including country, source, and topic fixed effects. If the diff-in-discs assumptions hold, then the LATE of the invasion in Equation 7 is identified by β_1 and the HLATE in Equation 7 is identified by θ_1 . For all specifications, we use HC1 heteroskedasticity-robust standard errors clustered at the source level. For robustness, we followed Toral, 2024 and also tested a local linear regression with a triangular kernel (Cattaneo et al., 2024) within the optimal bandwidth (Calonico et al., 2020). This has no substantive impact on the results. We also tested this analysis excluding our three NATO countries (Albania, Hungary, and Turkey), which had no effect on the results. Finally, we also tested a model where we omitted articles that GPT had identified as being not primarily Russian focused. This also had no substantive impact on results.

To analyze hypothesis 2A, that sentiment towards Russia would decline even in articles unrelated to military topics, we utilized the same analyses, but subset the articles to exclude those that our GPT model had identified the topic as military.

Results

Table 2 presents our diff-in-discs results. Model 1 demonstrates that sentiment towards Russia in all media became significantly more negative in response to the invasion of Ukraine. This represents a decline in sentiment of 0.15 ($p < .001$), which is a 0.27 standard deviation decline. Substantively, this means that the mean sentiment of articles fell from -0.15 pre-invasion to -0.30 post-invasion. However, sentiment in state-owned media declined significantly less than in independent media, falling by 0.10 standard deviations less ($p < .01$), which is nearly half of the total decline in sentiment. Models 2 shows that the results are not substantively or statistically altered by the use of the optimal bandwidth. Model 3 shows that this result is not driven by the inclusion of NATO allies. In support of hypothesis 2A, Model 4 shows that this result is also robust to the removal of all articles GPT coded as military. Finally, Model 5 shows removes both military related articles as well as filtering out articles that GPT coded as not primarily Russian-related. We utilize this filtering in order to ensure that articles that were overall negative in tone but only mentioned Russia in passing did not unduly influence the outcome. The effects in this model are substantively unaffected.

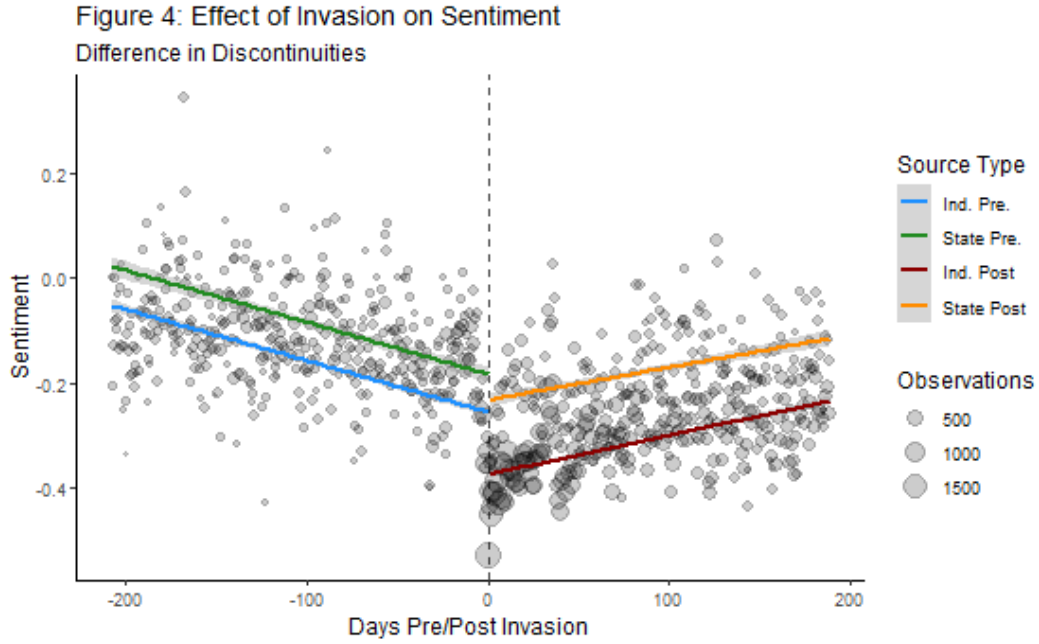
Figure 4 shows the heterogeneous effect of the invasion utilizing a linear model specification. This figure also supports the parallel trend assumption crucial to the causal identification of the diff-in-discs analysis. As can be seen, the sentiment of both independent and state-owned media show a similar downward trend towards the cutoff of the invasion. The diff-in-discs design identifies the HLATE by comparing the difference between the effect on state-owned media (here, the difference between the green line and the orange line) and on independent media (here, the difference between the blue line and the red line). Figure 5 shows the heterogeneous effect of the invasion on sentiment for non-military topics.

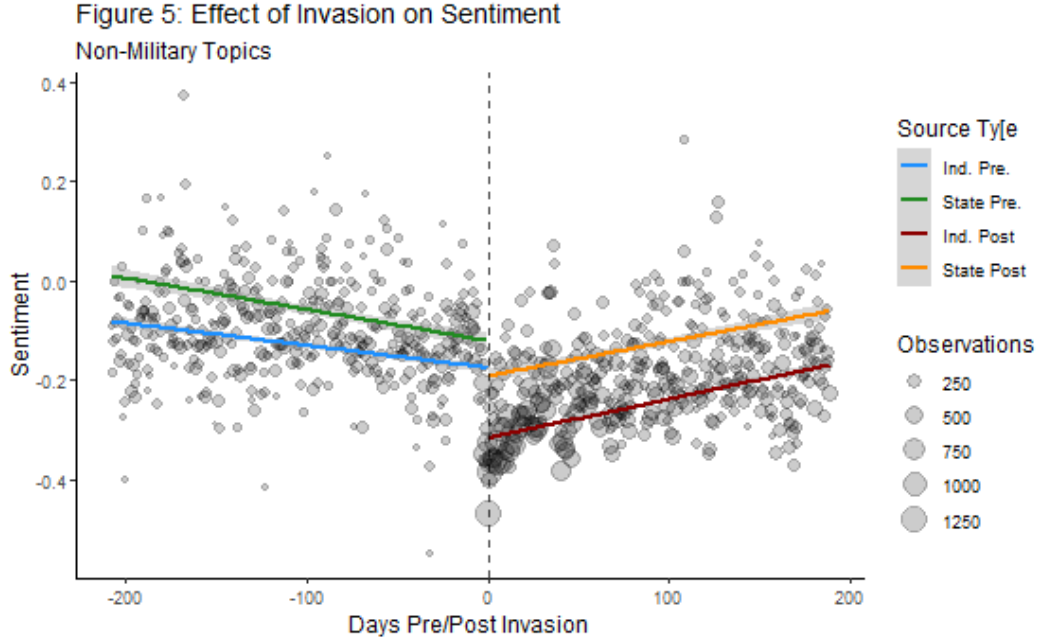
Table 2: Difference in Discontinuity Analysis

	(1)	(2)	(3)	(4)	(5)
Invasion	-0.15*** (0.01)	-0.191*** (0.013)	-0.158*** (0.014)	-0.14*** (0.01)	-0.176*** (0.013)
State-owned x Treatment	0.058** (0.018)	0.081** (0.025)	0.081** (0.025)	0.065*** (0.018)	0.095*** (0.023)
State FEs	X	X	X	X	X
Source FEs	X	X	X	X	X
Topic FEs	X	X	X	X	X
Num.Obs.	174094	62675	121894	130525	102765
R2	0.087	0.081	0.099	0.063	0.087
R2 Adj.	0.087	0.080	0.098	0.063	0.086

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Model 2 uses the triangular kernel within the optimal bandwidth. Model 3 omits NATO allies. Model 4 omits military-related articles. Model 5 omits military-related articles and only uses articles that were classified as primarily Russian Related.





In summary, these results demonstrate that independent media sources responded to the Russian invasion of Ukraine by significantly increasing negative sentiment towards Russia, and significantly more than state-owned sources do. This is true even in articles that do not directly cover military-related topics. This suggests that independent media, even in countries that are predisposed to be friendly toward Russia viewed the invasion negatively but that state-owned media responded differently. This aligns with our theory that state-owned media strategically controlled the sentiment towards Russia in its reporting after the invasion.

Mechanisms

Given the lack of agreement in the literature, we believe there are three potential mechanisms that could explain this discontinuity in media sentiment. The first, our argument, is a strategic decision by states in the Russian near abroad to signal their tethered relationship through the use of their state-owned media sources. In this case, state-owned media must balance between minimalist coverage—in the internet age, full censorship seems unlikely in our sample—of ongoing events with the state’s relationship with Russia. By restricting expressions of negative sentiment in reporting, state-owned

media signals support to Russia.

The first alternative mechanism might be that state-owned media is more fact based, with better access events as they happen and more funding. However, state-owned media also might fall prey to state control of the narrative (Djankov et al., 2003). We also know that in many of the countries of interest, media, political and business interests are often aligned, producing a skewed view of ongoing events (“Media Freedom”, 2023). Additionally, the volume of articles for state-owned media is significantly less and in the context of the invasion of Ukraine, much of the coverage was publicly available through social media channels. Therefore, we are unconvinced that this is the primary mechanism that explains this difference.

Finally, the second alternative mechanism is that state-owned and independent media covered topics at different rates, and it could be that our GPT model was more likely to code military events more negatively than other types of reporting. If this were the case, then different topic coverage would explain the results that we see in Figure 4. However, this mechanism would be insufficient to explain the results reported in Figure 5, which says that sentiment towards Russia changes significantly in non-military events.

Conclusion

In this article we have demonstrated that media coverage in countries in eastern Europe and central Asia responded to the invasion of Ukraine by increasing negative sentiment towards Russia in their reporting. We have shown that this increase in negative sentiment is not driven by an increase in coverage of military topics, but is true across a broad range of other topics. Utilizing a RDD design, we have shown that the invasion decreased average reported sentiment towards Russia by 0.17 points on a scale of -1 (negative) to 1 (positive), a 0.27 standard deviation decline. Additionally, we have shown that there is a differential effect between state-owned and independent media outlets. We utilized a diff-in-discs analysis to show that state-owned media sentiment toward Russia declined by 0.06 points less than independent sentiment, which

is roughly half of the reported sentiment change in independent media in that analysis. We have argued that the mechanism that explains this difference is strategic decision making by states, reflected in their management of state owned media. Theses smaller states, in close physical and geopolitical proximity to Russia, strategically restrict negative sentiment towards Russia in their own reporting. We provided support for this argument by highlighting that the difference is not explainable by either an assumption of higher factual reporting on behalf of state-media sources, nor by a difference in the topics that independent media and state-owned media are reporting on.

Given that this is one of the most significant territorial changes of the post Cold War order, this is an important point in our understanding of the role that military force still plays on the international stage. This work also suggests that some states are more strongly embedded in the international order than others. States that are less firmly embedded are not monoliths however; their independent media sources still suggest that leaders, journalists, or the public believes in the freedom of the press. This also has emphasizes the importance of a free and independent press as a key pillar of the liberal international order. More work must be done to understand the true implications of this divide. Most previous research suggests that elites shape the views of the mass public. But with an issue like contested territorial change, do elite views matter? Should we be concerned about the differing roles of state-owned and independent media? Future work can build off the work presented here to examine the role of state-owned media both as a tool for affecting public opinion, but also as a tool for signalling to the international system.

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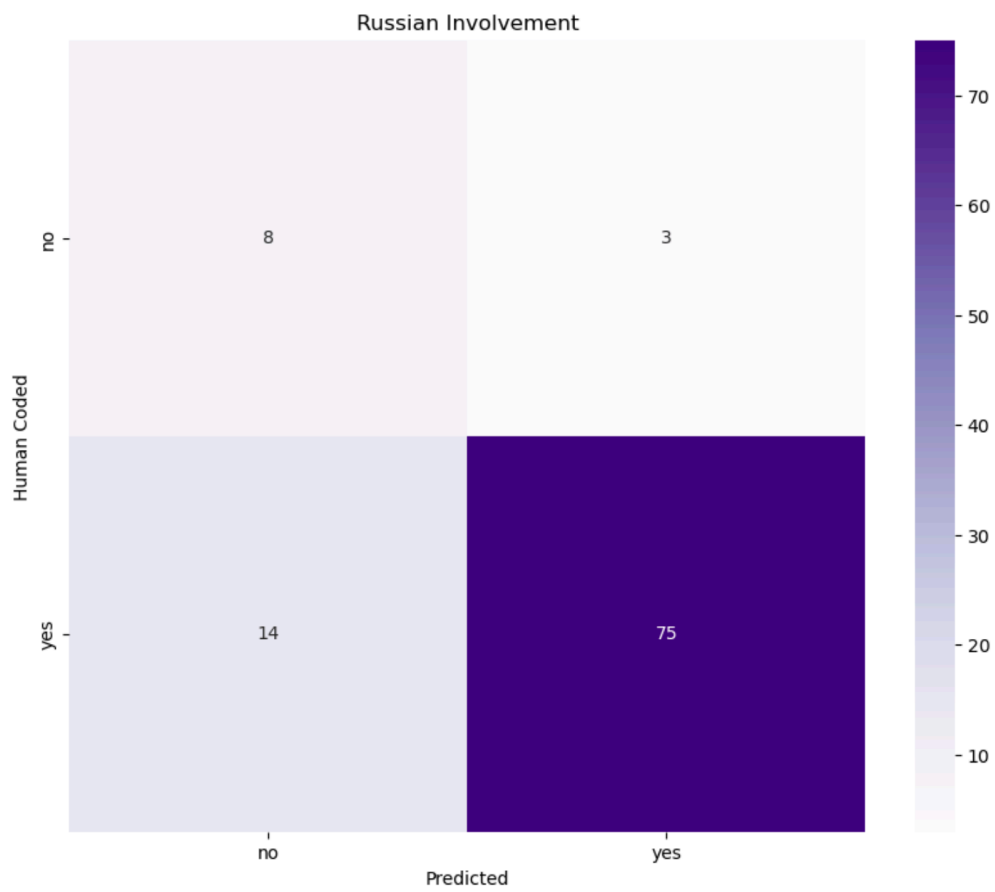
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Appendix



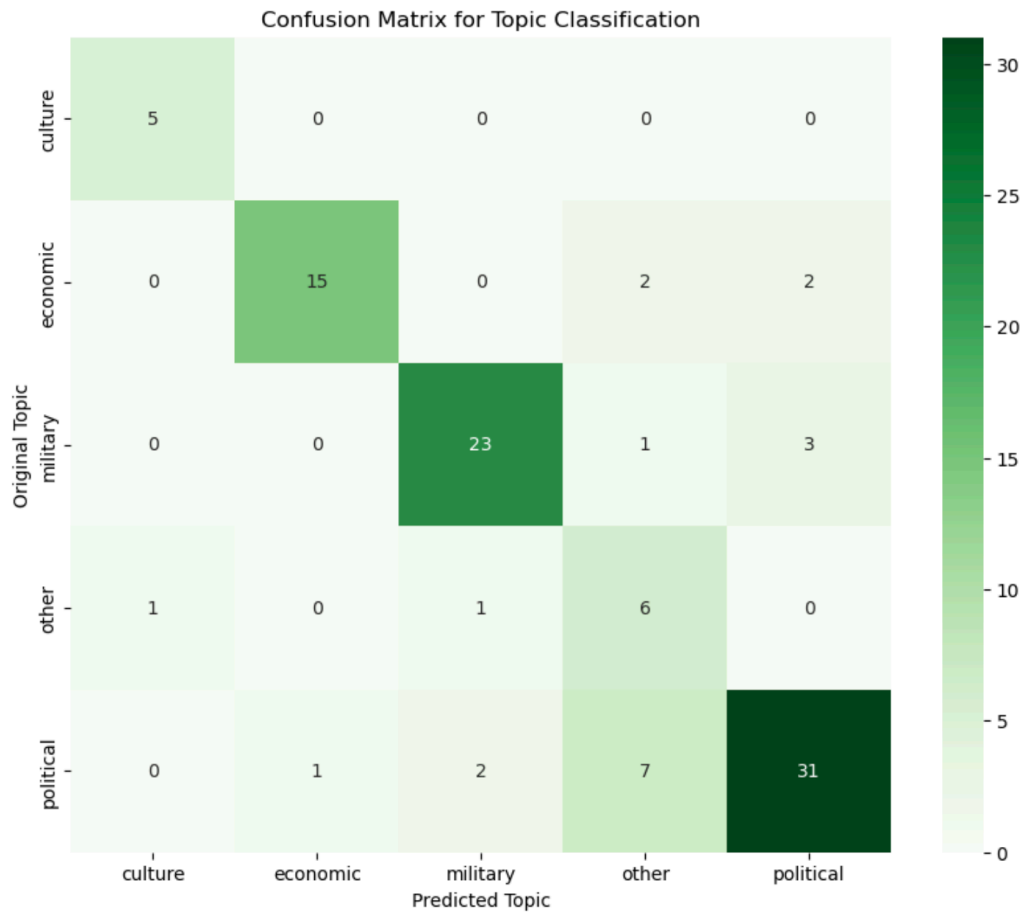


Figure 2 Appendix: Effect of Invasion on Sentiment
All Sources and Topics

